

MODEL QUESTION PAPER**IOT SECURITY****Time : 3 Hour****Max.Marks : 75****PART A****I. Answer all the following questions (9 x 1 = 9 Marks)**

1	What is the main difference between IoT and Machine-to-Machine (M2M) communication?	M 1.01	R
2	Which of the following is an application area of IoT? a) Virtual reality gaming b) Historical archaeology c) Energy generation d) Music composition	M 1.02	R
3	What is the purpose of an array in Arduino programming?	M 2.02	R
4	Which function is used to read digital input in Arduino?	M 2.03	A
5	Which communication module is commonly used for interfacing with Wi-Fi networks in Arduino?	M 3.05	U
6	Which module is commonly used for measuring acceleration in Arduino projects?	M 3.06	R
7	Which component is commonly used to display characters on an Arduino board?	M 3.03	R
8	What is the purpose of security architecture in the Internet of Things?	M 4.02	R
9	What type of attacks are specific to IoT?	M 4.04	R

PART B**II Answer any 8 questions from the following .Each questions carry 3 marks.****(8 x 3 = 24 Marks)**

1	Define the Internet of Things (IoT) and explain its key components.	M 1.01	U
2	Define sensors and discuss their characteristics and considerations in IoT applications	M 1.03	R
3	Explain the main components found on an Arduino board	M 2.01	U
4	Write an Arduino code snippet that uses a for loop to blink an LED connected to pin 13 five times with a 500ms delay.	M 2.04	A
5	What is an interrupt in Arduino programming?	M 2.06	U
6	Explain the concept of serial communication. What are the main components required for serial communication with Arduino?	M 3.01	R
7	Discuss the process of interfacing an SD card with Arduino using SPI communication.	M 3.06	U

8	What are the key security requirements in an IoT architecture?	M4.01	U
9	Describe the components of a secure IoT architecture?	M 4.02	U
10	Explain the concept of attack and fault trees in analyzing and mitigating security risks in IoT systems.	M4.06	U

PART C

Answer ALL questions. Each carry 7 marks.

(6 x 7 = 42 Marks)

1.	Compare and contrast IoT and M2M technologies, highlighting their similarities and differences. OR	M 1.01	U
2.	Explain the differences between scalar, multimedia, hybrid, and virtual sensors, providing examples of each	M 1.03	R
3.	Differentiate between microcontroller boards, single board controllers, and system-on-chip boards used in IoT applications.	M 1.05	U
4.	OR Define actuators and describe their types and characteristics in the context of IoT	M 1.04	U
5.	Explain the structure of an Arduino program, including the setup() and loop() functions. OR	M 2.02	U
6.	Discuss the process of connecting a button to an Arduino board and reading its state	M 2.05	R

7.	Explain the steps involved in setting up UART communication between Arduino and an ESP8266 module. What are the key considerations for successful communication?	M3.04	U
8.	O R Explain the process of interfacing an ADXL345 module with Arduino using I2C communication. What are the necessary connections and libraries?	M3.05	U
9.	Describe the process of interfacing an LCD character display with Arduino for parallel communication. What are the necessary connections and libraries?	M3.02	U
10.	OR Explain the steps involved in interfacing an Ethernet module with Arduino using SPI communication. What are the key considerations for successful communication?	M3.06	U
11.		M4.03	U
12.	.	M4.05	U